We use

- vertices_coordinate_mat $\begin{bmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} & \cdots & \begin{pmatrix} x_n \\ y_n \end{pmatrix} \end{bmatrix}$ (vertex) → we get it when creat mesh
- Cell_moments_array $Q[i,j,k] = \int \phi_i U_j$ in cell k
  - ϕ_i
  - U_j
  - cell k

in Cell_geo so we save them in

Struct DG States

---

init DG states:

After init mesh, we already have initial

vertices_coordinate_mat, mesh

Then we can build geo for each cell to compute init moment

for each K

$$Q_K = \begin{bmatrix} \int \phi_1 u_1 & & & \\ \vdots & & & \\ \int \phi_{nb} u_1 & & & \end{bmatrix} \quad (1, 2, 3, 4)$$

$$Q_{ijK} = \sum_l \phi_i(q_l) u_i(q_l) w_l$$

↓

each problem has init_state as given function

$$Q[:,:,K] = \sum_l w_l \phi_i(q_{il}) \begin{bmatrix} u_1(q_l) \\ \vdots \\ u_4(q_l) \end{bmatrix}$$

is given → We will define it later

basis_quad (i, l) (base, quad)

$$\begin{bmatrix} \phi_1(q_1) & & \\ \vdots & \cdots & \phi(q_l) \\ \phi_{nb}(q_1) & & \end{bmatrix}$$

01-04 for general ALE DG is done

Then we do Euler specifically!